

Air Quality Forecasting in New Delhi: A Hybrid Ensemble Machine Learning Approach

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Abstract—This project addresses the critical issue of air quality in New Delhi by developing a machine learning pipeline to forecast PM2.5 levels. Leveraging a heterogeneous dataset spanning 2019 to 2025, comprising air quality, meteorological, and solar radiation data, we engineered a robust feature set including lag features, rolling statistics, and cyclical time encodings. We evaluated multiple models, ranging from statistical baselines to advanced ensemble methods (XGBoost, CatBoost, LGBM). Our proposed solution is a Hybrid Ensemble Voting model combining CatBoost and XGBoost, which achieved the lowest MAE of 15.71. The model demonstrates strong capabilities in capturing seasonal trends and normal variances, though it remains conservative regarding extreme volatility. A Proof of Concept (PoC) web dashboard was developed to visualize predictions and provide health-related insights.

Index Terms—Machine Learning, Time Series Forecasting, PM2.5, Ensemble Learning, Air Quality, Urban Intelligence.

I. INTRODUCTION

Air pollution is one of the most pressing challenges in modern urban environments, with New Delhi consistently ranking among the most polluted cities globally. The primary pollutant of concern, Particulate Matter 2.5 (PM2.5), poses severe health risks to the population. Accurate forecasting of these levels is essential for public health planning and personal safety decisions.

This project falls under the theme of “Urban Intelligence & City Futures.” We aim to predict daily PM2.5 concentrations by integrating historical pollution data with weather and atmospheric parameters. Unlike simple statistical approaches, our work utilizes advanced machine learning techniques to capture the complex, non-linear dependencies between meteorological conditions and air quality.

II. PROBLEM DEFINITION

The objective is to forecast the daily average PM2.5 concentration ($\mu\text{g}/\text{m}^3$) for New Delhi. This is treated as a regression problem within a time-series context. The challenge lies in the high seasonality of the data and the presence of extreme outliers caused by specific events (e.g., crop burning, fireworks) which standard models often fail to capture. The solution targets stakeholders such as city planners, sensitive health groups (asthma patients), and the general public.

III. DATA ACQUISITION & PREPARATION

We curated a dataset from three distinct sources, covering the period from 2019 to 2025:

- 1) **OpenAQ:** PM2.5 data for New Delhi (curled from AWS S3 bucket) in CSV/GZ format [1].
- 2) **Visual Crossing:** Meteorological data (Temperature, Precip, Windspeed, etc.) via API [2].
- 3) **Copernicus Climate Data Store:** Solar Radiation and Boundary Layer Height data in .nc format [3].

A. Preprocessing

The raw data required significant cleaning. We removed impossible PM2.5 values (negative or distinct outliers). Since the air quality data was hourly, it was aggregated to daily averages. A major challenge was a gap of 109 days in the PM2.5 data. Standard interpolation was insufficient due to the gap size. We employed an imputation strategy based on historical yearly averages, leveraging the highly seasonal nature of the data.

The weather datasets were complete. Columns with high correlation or low relevance (based on domain knowledge) were dropped. The `conditions` column (categorical) was one-hot encoded as its cardinality was low (5 categories). Finally, the datasets were merged on the `date` index.

IV. FEATURE ENGINEERING

To enable the models to learn temporal dependencies, we introduced several engineering techniques:

- **Lag Features:** Lags of up to 7 days (Lag_1 to Lag_7) were created to capture immediate historical context.
- **Rolling Statistics:** Rolling mean and rolling standard deviation of the past 7 days were added to smooth noise and highlight trends.
- **Cyclical Encoding:** Day and Month features were transformed using Sine and Cosine functions to preserve their cyclical nature (e.g., December is close to January).
- **Calendar Features:** Day of week and Is Weekend flags were added to account for variations in human activity (traffic, industry).

V. DETAILED METHODOLOGY

A. Train-Test Split

The dataset spans 2019–2025. We utilized a temporal split:

- **Training:** 2019–2024 (6 years).
- **Testing:** 2025 (Jan–Nov).

B. Models Implemented

We implemented a pipeline ranging from baseline to hybrid ensembles:

- 1) **Baseline:** Simple Statistical Average.
- 2) **Classical ML:** Linear Regression, Support Vector Regressor (SVR), Elastic Net.
- 3) **Ensemble:** XGBoost, LightGBM (LGBM), CatBoost.
- 4) **Hybrid Ensemble:** A weighted voting approach combining our top two performers.

C. Hyperparameter Tuning

We performed extensive Grid Search for tuning:

- **SVR:** Tuned Kernel (rbf, poly, linear), C, Epsilon, and Gamma.
- **Elastic Net:** Tuned Alphas and L1_ratio.
- **CatBoost:** Tuned Depth, Learning Rate, Iterations, and L2 Leaf Reg.
- **LGBM:** Tuned Num Leaves, Learning Rate, N Estimators, and Max Depth.

The **Hybrid Ensemble** formula was derived empirically:

$$\text{Pred}_{\text{ensemble}} = (0.72 \times \text{Pred}_{\text{cat}}) + (0.28 \times \text{Pred}_{\text{xgb}}) \quad (1)$$

VI. RESULTS

The performance of all models on the 2025 test set is summarized in Table I. The Hybrid Ensemble achieved the best performance across all metrics.

TABLE I
MODEL PERFORMANCE METRICS (TEST SET 2025)

Model	MAE	RMSE	MSE	MAPE
Baseline	57.25	68.28	4661.86	1.36
Linear Regression	19.71	30.07	904.23	0.37
SVR	18.38	30.04	902.69	0.38
Elastic Net	19.41	29.67	880.42	0.36
XGBoost	16.33	26.40	697.13	0.22
CatBoost	15.95	25.73	662.13	0.22
Hybrid Ensemble	15.71	25.33	641.82	0.21

VII. ANALYSIS

A. Feature Importance Analysis

The hybrid model combines two distinct “views” of the data.

1. CatBoost (Temporal Focus): As shown in Fig. 1, CatBoost relies heavily on Autoregression. The `lag_1` feature is by far the most significant, indicating the model assumes “tomorrow looks like today.” It captures the temporal inertia effectively.

2. XGBoost (Physical Focus): Conversely, XGBoost prioritizes environmental drivers (Fig. 2). Solar Radiation and

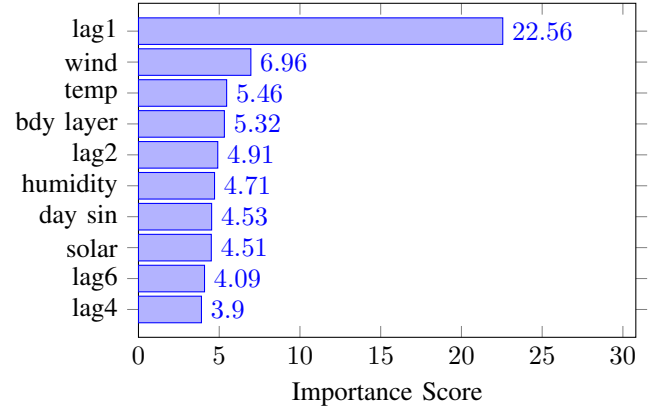


Fig. 1. CatBoost Top 10 Feature Importance: Dominance of Lag features indicates strong autoregressive behavior.

Boundary Layer Height are the primary split features. This captures the chemical/physical nature of the problem: solar radiation drives photochemical reactions, and boundary layer height determines pollutant trapping.

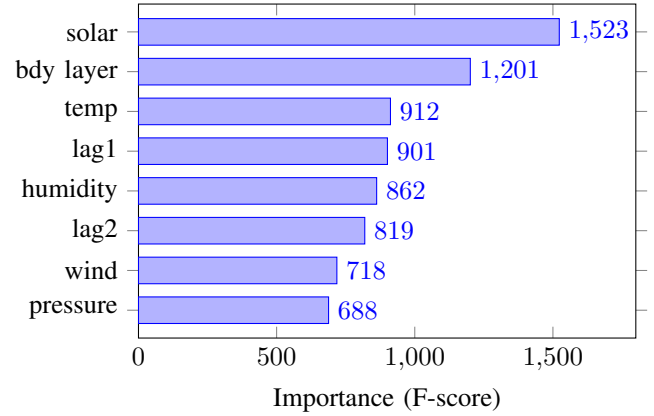


Fig. 2. XGBoost Top Feature Importance: Prioritizes physical environmental factors over history.

B. Model Evaluation

We analyzed the errors of the Hybrid model using three visualizations.

1. Trend Capture: Fig. 3 shows the model captures seasonality (lows in monsoon, highs in winter) effectively. However, it struggles with extreme volatility, smoothing out the highest jagged peaks.

2. Error Distribution: The KDE plot (Fig. 4) reveals a sharp peak at 0, indicating low bias. The model usually predicts very accurately, but the long tails confirm it is “bipolar”—mostly perfect, occasionally failing catastrophically on extreme days.

3. Stability (Residuals): The Residual plot (Fig. 5) displays heteroscedasticity. Errors are minimal in summer (May–Sep) but explode during high-pollution events (Nov–Jan). The massive spike in May likely corresponds to a specific dust storm event not captured by our features.

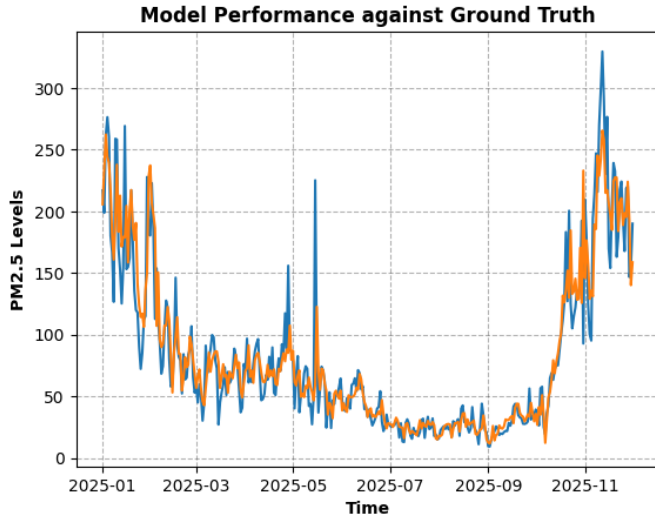


Fig. 3. Model Performance vs Ground Truth. The model tracks the trend well but misses extreme outliers.

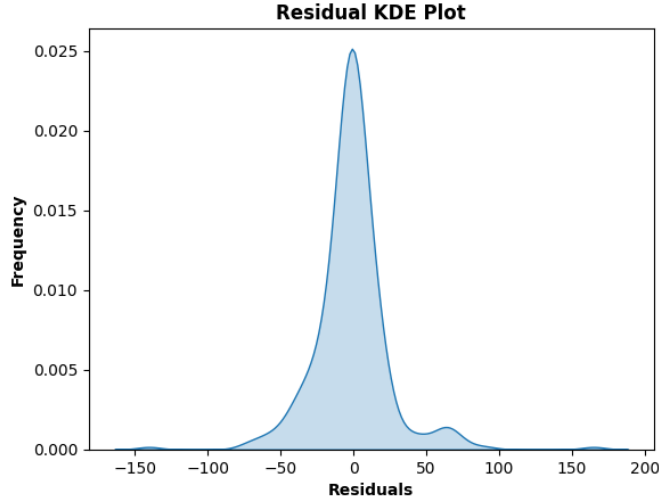


Fig. 4. Residual KDE Plot. The sharp peak at 0 indicates the model is unbiased and generally precise.

VIII. PROOF OF CONCEPT (POC)

We developed an interactive web dashboard using Streamlit.

- **Validation Section:** Allows users to visualize model performance against real 2025 values.
- **Prediction & Health Section:** Users can input custom weather/lag features to generate a PM2.5 forecast. The system provides actionable health insights:
 - Workout recommendations (Home vs. Outdoor).
 - School operation status.
 - “Cigarette equivalence” of breathing the air.
 - Mask mandates and Asthma sensitivity alerts.

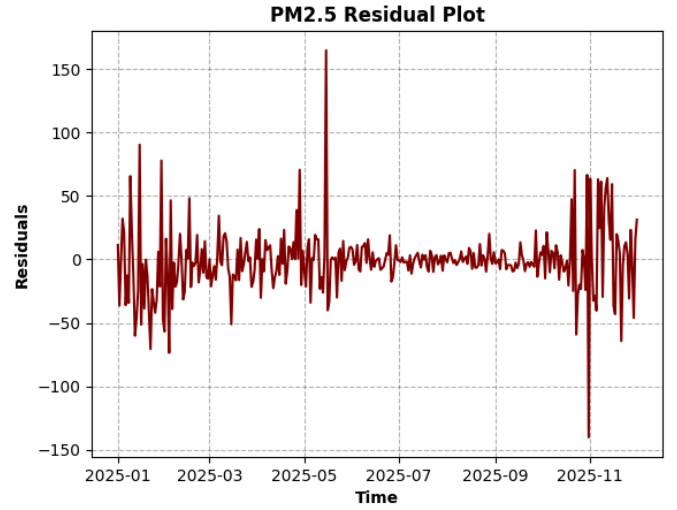


Fig. 5. Residual Plot over Time. Shows heteroscedasticity with higher errors in winter months.

IX. DISCUSSION

Our analysis suggests the model is statistically sound and unbiased. It understands “normal” air quality behavior and seasonal cycles perfectly. However, it is a “conservative” model. By minimizing RMSE, it plays it safe and predicts trends rather than risking extreme predictions for sudden spikes. We found that features like Precipitation and categorical Conditions were of low importance and could likely be dropped in future iterations.

X. FUTURE WORK

While the current ensemble approach establishes a robust baseline, there are several avenues for future enhancement:

- **Deep Learning Integration:** The current model struggles with extreme volatility. Future work could incorporate Long Short-Term Memory (LSTM) networks or Transformer-based architectures (e.g., TimeGPT) to better capture complex, non-linear temporal dependencies and sudden spikes.
- **External Data Sources:** The massive unexplained error spike in May suggests missing explanatory variables. Integrating real-time satellite data, such as Aerosol Optical Depth (AOD), or specific agricultural fire warnings could improve responsiveness to these acute events.
- **Quantile Regression:** Instead of a single point estimate, future models could utilize Quantile Regression to predict confidence intervals. This would quantify the uncertainty during high-risk winter months, allowing stakeholders to prepare for worst-case scenarios rather than just average trends.

XI. CONCLUSION

We successfully developed a Hybrid Ensemble model for PM2.5 forecasting in New Delhi. By combining the temporal

strengths of CatBoost with the physical modeling capabilities of XGBoost, we achieved a highly competitive MAE of 15.71. While the model struggles with extreme volatility, it provides a reliable baseline for daily urban planning and health advisory systems.

CODEBASE & RESOURCES

The complete, reproducible code and evaluation notebooks are available in our GitHub repository:

- **GitHub Repo:**

<https://github.com/abdullahhrashid/>

[air-quality-forecasting-using-classical-machine-learning](#)

A video demonstration of the Proof of Concept dashboard is available here:

- **Demo Video:**

[https://www.loom.com/share/](https://www.loom.com/share/237381417ab84d0a9540d8125bc6729f)

[237381417ab84d0a9540d8125bc6729f](https://www.loom.com/share/237381417ab84d0a9540d8125bc6729f)

REFERENCES

- [1] OpenAQ, "Open Air Quality Data," 2025. [Online]. Available: <https://openaq.org>.
- [2] Visual Crossing, "Weather Data Services," 2025. [Online]. Available: <https://www.visualcrossing.com>.
- [3] Copernicus Climate Change Service, "ERA5 hourly data on single levels from 1940 to present," 2025. [Online]. Available: <https://cds.climate.copernicus.eu>.